Options Volatility Modeling

Comparing Black-Scholes and SABR Models
on Real SPY Market Data

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Erdős Institute | Quant Finance Boot Camp, Spring 2026

OVERVIEW

What We Cover Today

01 The Problem

Black-Scholes and its constant volatility assumption

02 Market Evidence

The volatility smile from real SPY options data

03 The SABR Model

Stochastic volatility - parameters and intuition

04 Calibration

Fitting SABR to real market data

05 Key Results

90.78% improvement in pricing accuracy

06 Limitations

Where SABR still struggles and what comes next

THE PROBLEM

Black-Scholes: The Industry Standard

Key Assumptions

- Volatility is constant across all strikes
- Returns are log-normally distributed
- No jumps in asset price
- Frictionless, continuous markets

The Fatal Flaw

Real markets show that implied volatility varies across strikes → a pattern called the volatility smile or skew.

Black-Scholes predicts:

A flat, horizontal volatility line

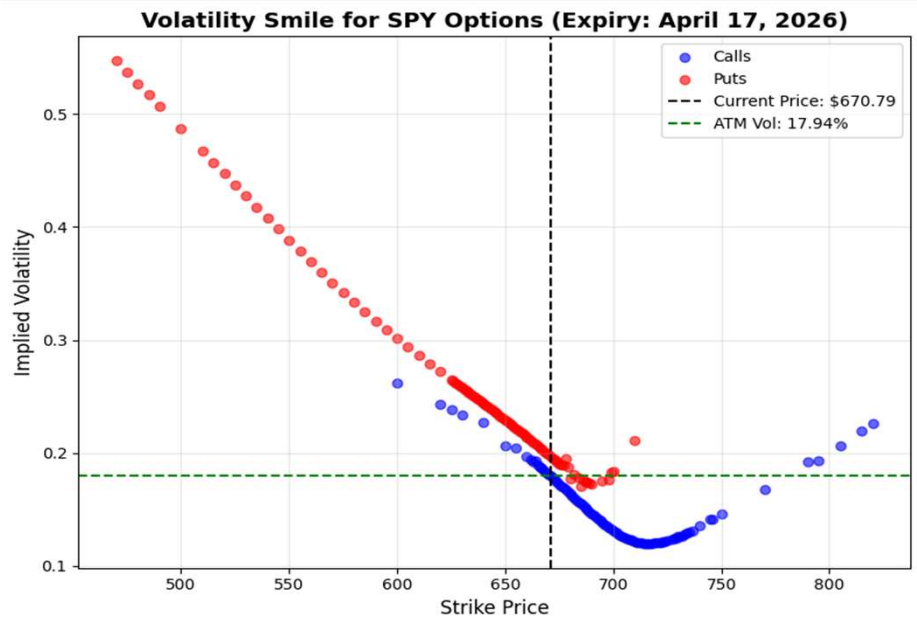
Reality shows:

A steep, asymmetric volatility skew

MARKET EVIDENCE

The Volatility Skew in Real SPY Data

SPY Options | Expiry: April 17, 2026 | Data collected: March 18, 2026 | **ATM Vol: 17.94%**



Puts (red) show high vol at low strikes
Calls (blue) show lower vol at high strikes

Fear Premium

Deep OTM puts carry 50%+ implied vol → institutional crash hedging drives this

ATM Vol: 17.94%

Black-Scholes uses this single number for ALL strikes → clearly inadequate

Steep Left Skew

Vol rises sharply for lower strikes → the market prices crash risk heavily

THE SOLUTION

The SABR Stochastic Volatility Model

Core Idea: Instead of fixing volatility as a constant, SABR lets volatility itself evolve randomly → and correlates it with the asset price.

α Alpha	β Beta	ρ Rho	ν Nu
Overall volatility level Shifts the entire smile up or down	Fixed at 0.5 Controls how volatility scales with price level	Correlation between price & volatility Negative → downward skew (equity markets)	Volatility of volatility Controls curvature of the smile

Calibrated Parameters: $\alpha = 4.751$ $\beta = 0.500$ (fixed) $\rho = -0.768$ $\nu = 2.901$

METHODOLOGY

Calibrating SABR to Market Data

1	2	3	4	5
Fetch Data	**Clean & Filter**	**Extract IV**	**Calibrate**	**Compare**
SPY options chain	Remove illiquid	Root finding	L-BFGS-B	RMSE

💡 Calibration is identical in concept to training a ML model: minimize a loss function over parameters using gradient-based optimization.

Loss Function: Sum of squared differences between SABR implied vols and market implied vols across all strikes

Optimizer: L-BFGS-B: Limited-memory BFGS with bounds → approximates Hessian matrix for fast convergence while keeping $\alpha>0$, $-1<\rho<1$, $\nu>0$

Alpha Initialization: Must set $\alpha = \text{ATM_vol} \times F^{(1-\beta)}$ → naïve $\alpha\approx0.18$ produces SABR vols of ~4% instead of ~18%; F:Forward Price

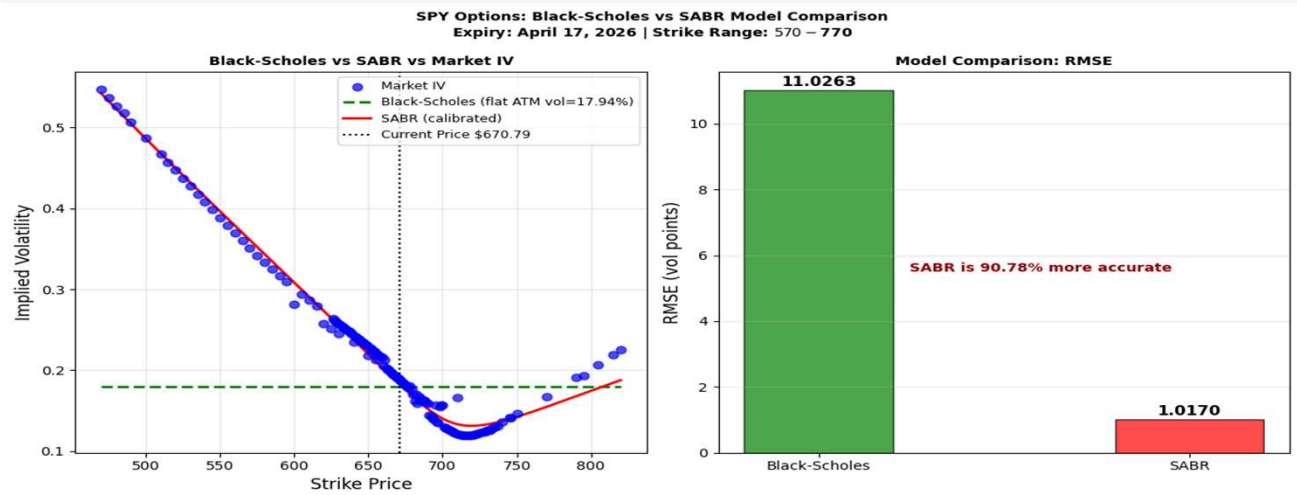
KEY RESULTS

SABR vs Black-Scholes: The Numbers

11.03
BS RMSE (vol points)

1.02
SABR RMSE (vol points)

90.78%
Accuracy Improvement



Left: Market IV (blue dots) vs Black-Scholes flat line (green) vs SABR fitted smile (red) | Right: RMSE bar chart

LIMITATIONS & FUTURE WORK

Where SABR Struggles and What Comes Next

⚠ Known Limitations

- Right-side misfit: SABR curves back up at high strikes while real equity volatility stays flat or dips
- Assumes symmetric smile shape → real equity skews are highly asymmetric
- Single expiry calibration only → full surface requires multi-expiry fitting
- Beta fixed at 0.5 by convention → cannot be reliably calibrated from a single snapshot

→ Future Directions

- SABR-LV (Local-Stochastic Vol): combines SABR with local volatility to fix the right-side fit
- Multi-expiry calibration: fit the full volatility surface across maturities
- Compare with Variance Gamma model → different philosophical approach to the smile
- Delta hedging analysis: compare hedging P&L under BS vs SABR assumptions

Key Takeaways

What This Project Shows

1. **Black-Scholes is inadequate**

   Constant volatility assumption fails visibly in real market data → the volatility skew is undeniable.

2. **SABR fixes the core problem**

   Stochastic, correlated volatility naturally reproduces the skew → 90.78% more accurate than BS.

3. **Calibration is the hard part**

   Getting parameters right requires correct initialization (α scaling), proper bounds, and the right strike range.

4. **Models have limits**

   SABR still struggles on the right side of the smile → no model is perfect and knowing why matters.